Aerial Autonomous Docking Control Method of Quadrotor UAV

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***Abstract*—The unmanned aerial vehicles (UAV) can autonomously assemble in the air, so as to deal with different scenarios and mission requirements. However, on the problem of autonomous aerial docking of rotorcraft UAVs, there are few existing research and most of them focus on docking with hovering UAVs. This paper proposes a solution to the problem of docking with hovering or moving quadrotor. The whole docking process is divided into the approach phase, the tracking phase and the docking phase. In the tracking phase and the docking phase, guidance methods are proposed according to the phase targets correspondingly. A docking controller which can adaptively estimate the desired point velocity and external disturbance is used to control the quadrotor UAV to complete the docking. The final simulation results show that the proposed method can complete the autonomous docking with a moving quadrotor UAV under disturbance.**

Keywords—quadrotor, autonomous docking, backstepping adaptive control, disturbance rejection, guidance

I. Introduction

In recent years, UAV technology has developed rapidly and been widely used. UAV has the advantages of small size, low cost and high safety. It can often be seen in civil, commercial, military and other fields. However, the load capacity of a single UAV is limited, so there are many restriction when it comes to performing tasks, such as the limited flight time and flight range due to the limited fuel or battery power. In addition, a single UAV has disadvantages such as fixed function and poor versatility. One of the most potential solutions is using aerial self-assembling UAV, which consists of multiple UAVs. Combination types are not limited to the combination of two rotor UAV[1], fixed wing UAV and rotor UAV[2], two fixed wing UAV[3], etc. UAVs can be assembled by themselves through the docking mechanism, which can improve the versatility, flexibility and load capacity of UAVs. This method can greatly improve the practical ability of UAV [4-5].

The research object of this paper is quadrotor UAV. Common control methods of this kind of UAV include backstepping control, active disturbance rejection control[6], model predictive control[7], adaptive control[8], sliding mode control[9] and combinations of different methods[10]. The most typical docking problem of UAV is the autonomous landing on mobile platforms[11] . For example, a backstepping adaptive controller was proposed in [12], and smooth landing was achieved by building safety ball and virtual point velocity dynamics. Reference [13] used discrete model predictive control and neural optimizer to achieve docking and landing with the mobile platform. Reference [14] proposed a rotor autonomous landing and tracking method using hybrid reinforcement learning strategy, which can achieve a higher success rate than traditional methods. There are relatively few existing researches on autonomous docking between rotor UAVs, but some research progress has been made: Reference [15] proposed a homing control law for VTOL quadrotor aircraft, it makes the aircraft move towards the moving target at a constant longitudinal velocity. Reference [16] realized aerial vertical combination of rotor UAVs through vision and tilt control law. Reference [17] proposed the concept of ModQuad UAV. The author completed aerial docking with a hovering UAV using gradient field function and velocity control. Based on ModQuad, data fusion of vision and accelerometer is used as navigation information in [18], and then the desired trajectory is generated and tracked by the controller to realize the autonomous assembly of UAV . However, the method is relatively simple and has no anti-disturbance ability.

This paper mainly deals with the aerial autonomous docking technology between quadrotor UAVs, which is also known as modular UAVs aerial autonomous docking technology. Existing researches on UAV docking mainly focus on platform landing, aerial refueling, etc. There are few researches on autonomous docking between rotor UAVs, and the existing research methods and scenarios are relatively simple, such as using a simple PID controller or keeping a UAV hovering.

In practice, sometimes UAVs need to dock when performing tasks between two places. Therefore, in this study, one UAV keeps hovering or flying at uniform speed, and the other UAV is controlled to dock actively. Firstly, some basic coordinate systems are defined for the docking problem of quadrotor UAVs and the mathematical model of the problem is given. Then, a backstepping adaptive controller for autonomous docking is designed, which can estimate the speed of trajectory points and external disturbance online to ensure that the UAV can track the desired trajectory. The docking process is divided into three different phases: the approximation phase, the tracking phase and the docking phase. In the latter two phases, the docking guidance method is designed respectively according to the aim and characteristics of the phase. Finally, the feasibility of the proposed autonomous docking method is verified by simulation.

II. Model of UAV Docking

To realize aerial docking of quadrotor UAVs, one UAV docks actively, while the other UAV docks passively by hovering or moving at uniform speed. When the states of the two UAVs meet certain conditions, the docking mechanism can rigidly connect the two UAVs together. All of the quadrotor UAVs here adopt the body structure of '×'. In addition, when two UAVs are close to each other in the docking process, the relative position is obtained by visual method to improve the docking accuracy. This section begins by introducing some basic definitions and models in docking problem.

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A. Basic Coordinate Systems and Relations

The basic diagram of the rotor UAV docking is shown in Fig. 1. The black arrow represents the thrust generated by rotors. Meanwhile, we define the following five coordinate systems:

1) Inertial coordinate system $\{\boldsymbol{I}\}$ *:* The world coordinate system.

2) UAV body coordinate system $\{\boldsymbol{B}\}$ *:* It is rigidly connected with the UAV, and its origin is located at the mass center of UAV. The x-axis is in the plane of longitudinal symmetry of the body and points towards the nose. The z-axis is perpendicular to the x-axis and points upwards in the plane of symmetry of the UAV. The y-axis is determined according to the right-handed rule; In addition, the body coordinate system of the active docking UAV is $\{\boldsymbol{B}_1\}$, and the coordinate system of the passive docking UAV is $\{\boldsymbol{B}_2\}$, they can also be respectively referred to as the active UAV and passive UAV.

3) Active docking UAV camera coordinate system $\{\boldsymbol{C}\}$ *:* Its origin is located in the optical center of the camera, the z-axis of the camera system is parallel to the main optical axis of the camera, the x-axis and the y-axis are respectively parallel to the horizontal and vertical directions of the imaging, wherein the x-axis is to the right and the y-axis points down.

4) Passive docking UAV tag coordinate system $\{\boldsymbol{G}\}$ *:* Its origin is located in the center of the tag, the z-axis is perpendicular to the plane and points out, the x-axis is perpendicular to the z-axis and points to the right side of the tag, and the y-axis is determined according to the right hand rule.

5) Coordinate system of docking point $\{\boldsymbol{D}\}$ *:* Its origin is the desired docking position $\boldsymbol{p}_{\mathrm{D}}$, located at a fuselage distance on the docking side of the passive docking UAV. It is also the position of the active UAV after the two UAVs are rigidly docked, and it moves with the passive UAV. In this coordinate system, the o-x-y plane is parallel to the horizontal plane, and the x-axis is consistent with the yaw direction of the passive UAV, which can be obtained by rotating $[\pi/2, \Psi_2, \pi/2]$ in the order of z-x-y according to euler angle. Where Ψ_2 is the yaw angle of the passive UAV.

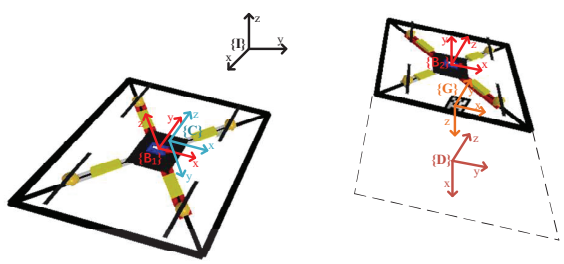

Fig. 1. Schematic diagram of coordinate systems.

In the docking process, we need to obtain the relative position of the docking point in active UAV. It is difficult to make the docking precision meet the requirements only by means of GPS or other sensors. Therefore, when the distance is close enough, the relative position is obtained by visual method. The accuracy of visual measurement can often reach the level of millimeter, and the closer the relative distance is, the higher the measurement precision will be. Autonomous docking does not require high precision at long relative distance, but high-precision data is required to meet the terminal requirements while UAVs are close, which is similar to vision. Assume that $\boldsymbol{d}_{tg}^{C}$ is the center position of the mark measured by the camera and the yaw angle Ψ_2 of the passive UAV is known, the rotation matrix from the inertial system to system $\{\boldsymbol{D}\}$ can be obtained:

$$\boldsymbol{R}_D^I = \boldsymbol{R}_z(\frac{\pi}{2})\boldsymbol{R}_x(\frac{\pi}{2})\boldsymbol{R}_y(\Psi_2) \tag{1}$$

Where $\boldsymbol{R}_z(\cdot)$, $\boldsymbol{R}_y(\cdot)$ and $\boldsymbol{R}_z(\cdot)$ represent the rotation matrix about the corresponding axis respectively. Position of the passive UAV $\boldsymbol{d}_{B_2}^{B_1}$ in coordinate system $\{\boldsymbol{B}_1\}$ is:

$$\boldsymbol{d}_{B_2}^{B_1} = \boldsymbol{R}_C^{B_1}\boldsymbol{d}_G^C + \boldsymbol{d}_C^{B_1} + (\boldsymbol{R}_{B_1}^I)^T\boldsymbol{R}_{B_2}^I\boldsymbol{R}_G^{B_2}\boldsymbol{d}_{B_2}^G \tag{2}$$

Further, the position of the docking point $\boldsymbol{d}_D^{B_a}$ under system $\{\boldsymbol{B}_1\}$ can be written as follows:

$$\boldsymbol{d}_D^{B_1} = \boldsymbol{d}_{B_2}^{B_1} + (\boldsymbol{R}_{B_1}^I)^T\boldsymbol{R}_D^I\boldsymbol{n} \tag{3}$$

Where $\boldsymbol{R}_{(\cdot)}^{(\cdot)}$ is the rotation matrix from the lower right coordinate system to the upper left coordinate system, $\boldsymbol{d}_C^{B_1}$ is the position of the camera in system $\{\boldsymbol{B}_1\}$, $\boldsymbol{d}_{B_2}^G$ is the position of the passive UAV in system $\{\boldsymbol{G}\}$, $\boldsymbol{n}$ is the vector from the mass center of the passive UAV to the docking point in system $\{\boldsymbol{D}\}$. Take Figure 1 as an example, the vector is $(0,0,l)^T$, l is the side length of the quadrotor UAV.

B. Mathematical Model of Quadrotor UAV

Considering the quadrotor UAV as a rigid body, the UAV dynamics model can be easily obtained as follows:

$$\begin{cases} \dot{\boldsymbol{p}} = \boldsymbol{v} \\ \dot{\boldsymbol{v}} = \boldsymbol{g} + \dfrac{1}{m}\boldsymbol{R}_B^I\boldsymbol{F} + \boldsymbol{f}_\varepsilon \\ \dot{\boldsymbol{\eta}} = \boldsymbol{H}(\boldsymbol{\eta})\boldsymbol{\omega} \\ \boldsymbol{J}\dot{\boldsymbol{\omega}} = \boldsymbol{M} - \boldsymbol{\omega}\times\boldsymbol{J}\boldsymbol{\omega} + \boldsymbol{\tau}_\varepsilon \end{cases} \tag{4}$$

Where, $\boldsymbol{p} = (x, y, z)^T$ is the position of the UAV under inertial frame, $\boldsymbol{v} = (v_x, v_y, v_z)^T$ is the velocity of the UAV under inertial frame, $\boldsymbol{g}$ is the gravity acceleration vector, $\boldsymbol{F} = (0,0,T)^T$ is the thrust generated by rotors in the UAV body system, $\boldsymbol{\eta} = (\phi, \theta, \Psi)^T$ is the euler angle of the UAV, rotating in order z-y-x, $\boldsymbol{\omega} = (p, q, r)^T$ is the angular velocity under the UAV body system, $\boldsymbol{J}$ is the inertia tensor, $\boldsymbol{M} = (M_x, M_y, M_z)^T$ is the torque generated by rotors. $\boldsymbol{f}_\varepsilon$ and $\boldsymbol{\tau}_\varepsilon$ are bounded unknown disturbance and are constant or vary slowly over time. $\boldsymbol{R}_B^I$ is the rotation matrix of inertia frame to body system and can be expressed as:

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$$\boldsymbol{R}_B^I = \begin{bmatrix} C_\theta C_\psi & S_\theta C_\psi S_\phi - S_\psi C_\phi & S_\theta C_\psi C_\phi + S_\psi S_\phi \\ C_\theta S_\psi & S_\theta S_\psi S_\phi + C_\psi C_\phi & S_\theta S_\psi C_\phi - C_\psi S_\phi \\ -S_\theta & C_\theta S_\phi & C_\theta C_\phi \end{bmatrix} \quad (5)$$

Where $C_{(\cdot)}$ is equivalent to $\cos(\cdot)$, $S_{(\cdot)}$ is equivalent to $\sin(\cdot)$, $\boldsymbol{H}(\boldsymbol{\eta})$ is the transformation matrix, which can be approximated as an identity matrix under small angle assumption.

Suppose the angular velocity of the four rotors are ω_1, ω_2, ω_3 and ω_4 respectively. The distance from the rotor to the mass center of the UAV is L, while the lift coefficient and reverse torque coefficient are C_T and C_M. The relationship between the control force and torque of the '×' type UAV and the rotor speed is:

$$\begin{cases} T = C_T(\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2) \\ M_x = \dfrac{\sqrt{2}}{2} C_T L(-\omega_1^2 - \omega_2^2 + \omega_3^2 + \omega_4^2) \\ M_y = \dfrac{\sqrt{2}}{2} C_T L(-\omega_1^2 + \omega_2^2 + \omega_3^2 - \omega_4^2) \\ M_z = C_M(-\omega_1^2 + \omega_2^2 - \omega_3^2 + \omega_4^2) \end{cases} \quad (6)$$

C. Condition of Docking

The two UAVs can be rigidly connected through the docking mechanism installed on the UAV, such as magnetic suction mechanism and mechanical mechanism, etc. This paper focuses on the control method of autonomous docking and does not elaborate too much on the docking mechanism. When the active docking UAV reaches the docking point and both UAVs' attitude is similar, the docking mechanism can rigidly connect the two UAVs to complete the docking. Different docking mechanisms have different requirements for terminal state error, and the velocity difference and angular velocity difference between the two UAVs should not be too large, otherwise it will bring adverse effects on the docking mechanism and subsequent control. Here, it is assumed that the docking state conditions is:

$$\begin{cases} \|\Delta \boldsymbol{p}\|_2 \le 0.05m \\ \|\Delta v\|_2 \le 0.2m/s \\ \|\Delta \boldsymbol{\eta}\|_\infty \le 0.05rad \\ \|\Delta \boldsymbol{\omega}\|_2 \le 0.1rad/s \end{cases} \quad (7)$$

Where, $\|\cdot\|_2$ represents the binary norm of the vector, $\|\cdot\|_\infty$ represents the infinite norm of the vector, $\Delta \boldsymbol{p}$ is the position difference between the active docking UAV and the docking point $\boldsymbol{p}_D$, and Δv, $\Delta \boldsymbol{\eta}$, $\Delta \boldsymbol{\omega}$ are the velocity difference, attitude angle difference and angular velocity difference of the two UAVs respectively.

III. Design of Docking Controller

In this section, we design a quadrotor UAV controller specially for docking, and calculate the control thrust T and control torque $\boldsymbol{M}$ according to the given desired state. Assuming that the external disturbance is bounded, the quadrotor UAV control system is divided into position subsystem and attitude subsystem. In the position subsystem, backstepping adaptive control is used to track the trajectory points, and the desired trajectory speed and external disturbance are estimated online by adaptive control. In the attitude subsystem, backstepping sliding mode control is used to reach the desired attitude angle, and the external disturbance is compensated using an adaptive method.

A. Postion Controller

The position controller calculates the thrust, desired pitch angle and desired roll angle of the UAV according to desired position. The expected tracking position is assumed to be $\boldsymbol{p}_\mathrm{d}$, and the position tracking error is defined as:

$$\boldsymbol{e}_p = \boldsymbol{p}_\mathrm{d} - \boldsymbol{p} \quad (8)$$

Take the derivative of the desired trajectory point as $\dot{\boldsymbol{p}}_\mathrm{d} = \boldsymbol{\Theta}$, set its estimated value as $\hat{\boldsymbol{\Theta}}$, and define its estimation error as $\Delta\boldsymbol{\Theta} = \hat{\boldsymbol{\Theta}} - \boldsymbol{\Theta}$. Define the Lyapunov function as:

$$V_1 = \frac{1}{2}\boldsymbol{e}_p^T \boldsymbol{e}_p + \frac{1}{2}\Delta\boldsymbol{\Theta}^T \boldsymbol{\Gamma}^{-1} \Delta\boldsymbol{\Theta} \quad (9)$$

Where $\boldsymbol{\Gamma} \in \mathbb{R}^{3\times3}$ is the adaptive rate. Since the motion of the docking target is static or uniform, it can be assumed that $\dot{\boldsymbol{\Theta}}$ is close to zero.The derivative of the above equation is:

$$\dot{V}_1 = \boldsymbol{e}_p^T(\boldsymbol{\Theta} - \boldsymbol{v}) + \Delta\boldsymbol{\Theta}^T \boldsymbol{\Gamma}^{-1} \dot{\hat{\boldsymbol{\Theta}}} \quad (10)$$

Then the speed virtual control quantity can be designed as:

$$\boldsymbol{v}^* = K_1 \boldsymbol{e}_p + \hat{\boldsymbol{\Theta}} \quad (11)$$

$$\dot{\hat{\boldsymbol{\Theta}}} = \boldsymbol{\Gamma} \boldsymbol{e}_p \quad (12)$$

Where, $\boldsymbol{K}_1 \in \mathbb{R}^{3\times3}$ is the parameter to be adjusted, and it is a positive definite diagonal matrix, we can get:

$$\dot{V}_1 = -\boldsymbol{K}_1 \boldsymbol{e}_p^T \boldsymbol{e}_p \quad (13)$$

The velocity error is defined as:

$$\boldsymbol{e}_v = \boldsymbol{v}^* - \boldsymbol{v} \quad (14)$$

It is equivalent to

$$\boldsymbol{v} = \boldsymbol{K}_1 \boldsymbol{e}_p + \hat{\boldsymbol{\Theta}} - \boldsymbol{e}_v \quad (15)$$

Then we have:

$$\dot{V}_1 = -\boldsymbol{e}_p^T \boldsymbol{K}_1 \boldsymbol{e}_p + \boldsymbol{e}_p^T \boldsymbol{e}_v \quad (16)$$

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The unknown variable is $\boldsymbol{\alpha} = -\boldsymbol{K}_1 \Delta\boldsymbol{\Theta} - \boldsymbol{f}_\varepsilon$, and its estimated value is $\hat{\boldsymbol{\alpha}}$. The estimation error is defined as $\Delta\boldsymbol{\alpha} = \hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}$. and the Lyapunov function is defined as follows:

$$V_2 = V_1 + \frac{1}{2}\boldsymbol{e}_v^T \boldsymbol{e}_v + \frac{1}{2}\Delta\boldsymbol{\alpha}^T \Lambda_\alpha^{-1} \Delta\boldsymbol{\alpha} \tag{17}$$

The derivative of $\boldsymbol{\alpha}$ is: $\dot{\boldsymbol{\alpha}} = -\boldsymbol{K}_1(\dot{\hat{\boldsymbol{\Theta}}} - \dot{\boldsymbol{\Theta}}) - \dot{\boldsymbol{f}}_\varepsilon = -\boldsymbol{K}_1 \boldsymbol{\Gamma} \boldsymbol{e}_p$. The derivative of $\boldsymbol{e}_p$ is: $\dot{\boldsymbol{e}}_p = \dot{\boldsymbol{p}}_d - \boldsymbol{v} = -\boldsymbol{K}_1 \boldsymbol{e}_p + \boldsymbol{e}_v - \Delta\boldsymbol{\Theta}$ .Take the derivative of (17) and we get:

$$\begin{aligned}
\dot{V}_2 &= \dot{V}_1 + \boldsymbol{e}_v^T (\boldsymbol{K}_1 \dot{\boldsymbol{e}}_p + \boldsymbol{\Gamma} \boldsymbol{e}_p - \boldsymbol{f}_\varepsilon - \boldsymbol{g} - \boldsymbol{a}_r) \\
&\quad + \Delta\boldsymbol{\alpha}^T \Lambda_\alpha^{-1} (\dot{\hat{\boldsymbol{\alpha}}} - \dot{\boldsymbol{\alpha}}) \\
&= -\boldsymbol{e}_p^T \boldsymbol{K}_1 \boldsymbol{e}_p + \boldsymbol{e}_v^T (\boldsymbol{e}_p - \boldsymbol{K}_1 \boldsymbol{K}_1 \boldsymbol{e}_p + \boldsymbol{K}_1 \boldsymbol{e}_v \\
&\quad + \boldsymbol{\Gamma} \boldsymbol{e}_p - \boldsymbol{K}_1 \Delta\boldsymbol{\Theta} - \boldsymbol{f}_\varepsilon - \boldsymbol{g} - \boldsymbol{a}_r) \\
&\quad + \Delta\boldsymbol{\alpha}^T \Lambda_\alpha^{-1} (\dot{\hat{\boldsymbol{\alpha}}} + \boldsymbol{K}_1 \boldsymbol{\Gamma} \boldsymbol{e}_p) \\
&= -\boldsymbol{e}_p^T \boldsymbol{K}_1 \boldsymbol{e}_p + \boldsymbol{e}_v^T ((\boldsymbol{I} - \boldsymbol{K}_1 \boldsymbol{K}_1 + \boldsymbol{\Gamma}) \boldsymbol{e}_p + \boldsymbol{K}_1 \boldsymbol{e}_v \\
&\quad - \boldsymbol{g} + \boldsymbol{\alpha} - \boldsymbol{a}_r) + \Delta\boldsymbol{\alpha}^T \Lambda_\alpha^{-1} (\dot{\hat{\boldsymbol{\alpha}}} + \boldsymbol{K}_1 \boldsymbol{\Gamma} \boldsymbol{e}_p)
\end{aligned} \tag{18}$$

Where $\boldsymbol{a}_r = \frac{1}{m} \boldsymbol{R}_B^I \boldsymbol{F}$ is a control variable to be designed, let:

$$\boldsymbol{a}_r = (\boldsymbol{I} - \boldsymbol{K}_1 \boldsymbol{K}_1 + \boldsymbol{\Gamma}) \boldsymbol{e}_p + (\boldsymbol{K}_1 + \boldsymbol{K}_2) \boldsymbol{e}_v - \boldsymbol{g} + \hat{\boldsymbol{\alpha}} \tag{19}$$

$$\dot{\hat{\boldsymbol{\alpha}}} = \Lambda_\alpha \boldsymbol{e}_v - \boldsymbol{K}_1 \boldsymbol{\Gamma} \boldsymbol{e}_p \tag{20}$$

Where $\boldsymbol{K}_2$ is a parameter to be adjusted. Substituting (19) and (20) into (18), we can get:

$$\dot{V}_2 = -\boldsymbol{e}_p^T \boldsymbol{K}_1 \boldsymbol{e}_p - \boldsymbol{e}_v^T \boldsymbol{K}_2 \boldsymbol{e}_v \le 0 \tag{21}$$

Let $\boldsymbol{a}_r = (a_x, a_y, a_z)^T$, according to (1) and (2), the thrust, desired roll angle and desired pitch angle can be expressed as:

$$\begin{cases}
T = \dfrac{m a_z}{\cos\theta \cos\varphi} \\
\varphi_d = \arcsin\left(\dfrac{m a_x \sin\psi - m a_y \cos\psi}{T}\right) \\
\theta_d = \arcsin\left(\dfrac{m a_x \cos\psi + m a_y \sin\psi}{T}\right)
\end{cases} \tag{22}$$

B. Attitude Controller

Set the desired attitude angle as $\boldsymbol{\eta}_{\mathrm{d}}$, in which the desired pitch angle and roll angle are given by the position controller, and the desired yaw angle is determined according to the mission. The desired yaw angle of the two UAVs is consistent. Considering that docking has requirements on attitude, assume the attitude of the passive docking UAV is $\boldsymbol{\eta}_2$, then define the attitude error:

$$\boldsymbol{e}_\eta = (1-\rho)(\boldsymbol{\eta}_{\mathrm{d}} - \boldsymbol{\eta}) + \rho(\boldsymbol{\eta}_2 - \boldsymbol{\eta}) \tag{23}$$

Where $\rho \in [0,1)$ is the weight coefficient. ρ is zero when not in the docking stage. When entering the docking section $\rho = \dfrac{\kappa_1}{\sqrt{1 + \kappa_2 \|\Delta\boldsymbol{p}\|^2}}$, κ_1 and κ_2 are adjustable parameters. The purpose is to adjust the relative position while adaptively reducing the attitude difference between UAVs at the end of docking stage.

Define the Lyapunov function as:

$$V_3 = \frac{1}{2}\boldsymbol{e}_\eta^T \boldsymbol{e}_\eta \tag{24}$$

Let the generalized desired attitude angle be $\boldsymbol{\eta}_{\mathrm{d'}} = (1-\rho)\boldsymbol{\eta}_{\mathrm{d}} + \rho\boldsymbol{\eta}_2$, and the derivative can be calculated by:

$$\dot{V}_3 = \boldsymbol{e}_\eta^T (\dot{\boldsymbol{\eta}}_{\mathrm{d'}} - \boldsymbol{\omega}) \tag{25}$$

Design angular velocity virtual control quantity as:

$$\boldsymbol{\omega}^* = \dot{\boldsymbol{\eta}}_{\mathrm{d'}} + \boldsymbol{K}_3 \boldsymbol{e}_\eta \tag{26}$$

Where $\boldsymbol{K}_3 \in \mathbb{R}^{3\times3}$ is a parameter to be adjusted.

Define angular velocity error as:

$$\boldsymbol{e}_\omega = \boldsymbol{\omega}^* - \boldsymbol{\omega} \tag{27}$$

Then:

$$\dot{V}_3 = -\boldsymbol{e}_\eta^T \boldsymbol{K}_3 \boldsymbol{e}_\eta + \boldsymbol{e}_\eta^T \boldsymbol{e}_\omega \tag{28}$$

Define the sliding mode surface as $\boldsymbol{s}$, where $\boldsymbol{A}$ is a undetermined symmetric positive definite matrix:

$$\boldsymbol{s} = \boldsymbol{A} \boldsymbol{e}_\eta + \boldsymbol{e}_\omega \tag{29}$$

Define a new Lyapunov function as:

$$V_4 = V_3 + \frac{1}{2}\boldsymbol{s}^T \boldsymbol{s} + \frac{1}{2}\Delta\boldsymbol{\tau}_\varepsilon^T \Lambda_\tau^{-1} \Delta\boldsymbol{\tau}_\varepsilon \tag{30}$$

Where Λ_τ is a symmetric positive definite matrix, and the derivative of (30) can be obtained:

$$\begin{aligned}
\dot{V}_4 &= -\boldsymbol{e}_\eta^T (\boldsymbol{A} + \boldsymbol{K}_3) \boldsymbol{e}_\eta + \boldsymbol{s}^T (\boldsymbol{e}_\eta + (\boldsymbol{A} + \boldsymbol{K}_3) \dot{\boldsymbol{e}}_\eta \\
&\quad + \ddot{\boldsymbol{\eta}}_d - \boldsymbol{J}^{-1} \boldsymbol{M} + \boldsymbol{J}^{-1} \boldsymbol{\omega} \times \boldsymbol{J}\boldsymbol{\omega} - \boldsymbol{J}^{-1} \boldsymbol{\tau}_\varepsilon) + \Delta\boldsymbol{\tau}_\varepsilon^T \Lambda_\tau^{-1} \dot{\hat{\boldsymbol{\tau}}}_\varepsilon
\end{aligned} \tag{31}$$

Given $\dot{\boldsymbol{e}}_\eta = \dot{\boldsymbol{\eta}}_{\mathrm{d'}} - \boldsymbol{\omega} = \dot{\boldsymbol{\eta}}_{\mathrm{d'}} - \boldsymbol{\omega}^* + \boldsymbol{e}_\omega = \boldsymbol{e}_\omega - \boldsymbol{K}_3 \boldsymbol{e}_\eta$, let:

$$\begin{aligned}
\boldsymbol{M} &= \boldsymbol{K}_4 \boldsymbol{J}\boldsymbol{s} + \varepsilon \boldsymbol{J} Sat(\boldsymbol{s}) + \boldsymbol{J}(\boldsymbol{I} - (\boldsymbol{A} + \boldsymbol{K}_3)\boldsymbol{K}_3) \boldsymbol{e}_\eta \\
&\quad + \boldsymbol{J}(\boldsymbol{A} + \boldsymbol{K}_3) \boldsymbol{e}_\omega + \boldsymbol{J}\ddot{\boldsymbol{\eta}}_{\mathrm{d'}} + \boldsymbol{\omega} \times \boldsymbol{J}\boldsymbol{\omega} - \hat{\boldsymbol{\tau}}_\varepsilon
\end{aligned} \tag{32}$$

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$$\dot{\hat{\boldsymbol{\tau}}}_{\varepsilon} = -\boldsymbol{\Lambda}_{\tau}\boldsymbol{J}^{-1}\boldsymbol{s} \tag{33}$$

Where, $\ddot{\boldsymbol{\eta}}_{d'}$ can be set as zero, $\boldsymbol{K}_4$ is a parameter to be adjusted, $Sat(\boldsymbol{s}) = (sat(s_1), sat(s_2), sat(s_3))^T$, and $sat(\cdot)$ is the saturation function, which can inhibit chattering:

$$sat(x) = \begin{cases} 1, x > \Delta \\ \dfrac{1}{\Delta}x, |x| < \Delta \\ -1, x < -\Delta \end{cases} \tag{34}$$

Substituting (32) and (33) into (31) gives:

$$\dot{V}_4 = -\boldsymbol{e}_{\eta}^T(\boldsymbol{K}_3 + \boldsymbol{A})\boldsymbol{e}_{\eta} - \boldsymbol{s}^T\boldsymbol{K}_4\boldsymbol{s} - \varepsilon\boldsymbol{s}^T Sat(\boldsymbol{s}) \le 0 \tag{35}$$

The flow chart of the docking controller is shown in Fig. 2.

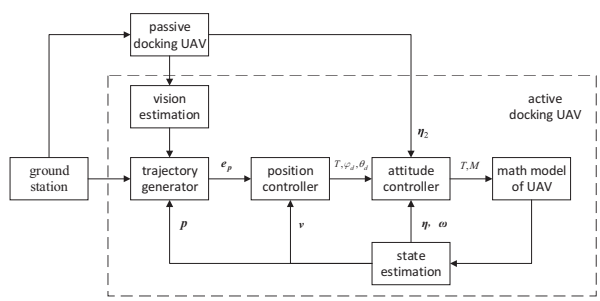

Fig. 2. Graph of docking controller.

IV. Guidance Method of Dockingd

In addition to a controller with better performance, a reasonable docking strategy and guidance method are also needed to complete the autonomous docking. Here, we divide the whole docking process into three phases: the approximation phase, the tracking phase and the docking phase.

A. Approximation Phase

The passive UAV keeps hovering or flies at a constant speed according to the task, while keeping the yaw direction consistent with the speed direction. The active UAV approaches the docking side of the passive UAV and navigates through GPS until the tag can be recognized. At this time, the tracking phase is entered.

B. Tracking Phase

At this phase, the active UAV should fly to a certain distance on docking side of the passive UAV, and then tracks the docking point for a certain time; The point $(0,0,d)$ in the coordinate system $\{\boldsymbol{D}\}$ is regarded as the target tracking point $\boldsymbol{p}_T$, and system $\{\boldsymbol{D}\}$ is regarded as a spherical coordinate system (r,α,β), where r is the distance from the coordinate point to the origin, α is the angle between the line from the coordinate point to the origin and the positive z-axis, β is the azimuth angle between the projection of the line from the coordinate point to the origin on the x-o-y plane and the positive x-axis. The spherical coordinate of the current position of active UAV is (r_1,α_1,β_1) . The spherical coordinate of the desired position is (r_d,α_d,β_d), its initial value is the spherical coordinate of the active docking UAV when it enters the tracking phase. The spherical coordinate of the tracking point is $(d,0,\pi)$, among which the third item π is set to ensure that the active docking UAV approaches from the rear of the docked UAV. The define of error is as follows:

$$\begin{cases} e_{r_{\mathrm{d}}} = d - r_{\mathrm{d}} \\ e_{\alpha_{\mathrm{d}}} = -\alpha_{\mathrm{d}} \\ e_{\beta_{\mathrm{d}}} = \pi - \beta_{\mathrm{d}} \end{cases} \tag{36}$$

Referring to the method in [12], the spherical coordinate derivative of the desired position is designed as:

$$\begin{cases} \dot{r}_{\mathrm{d}} = \dfrac{2e^{-c_1|r_1 - r_{\mathrm{d}}|}}{1+e^{-c_1|r_1 - r_{\mathrm{d}}|}}k_r e_{r_{\mathrm{d}}} \\ \dot{\alpha}_{\mathrm{d}} = \dfrac{2e^{-c_2|\alpha_1 - \alpha_{\mathrm{d}}|}}{1+e^{-c_2|\alpha_1 - \alpha_{\mathrm{d}}|}}k_\alpha e_{\alpha_{\mathrm{d}}} \\ \dot{\beta}_{\mathrm{d}} = \dfrac{2e^{-c_3|\beta_1 - \beta_{\mathrm{d}}|}}{1+e^{-c_3|\beta_1 - \beta_{\mathrm{d}}|}}k_\beta e_{\beta_{\mathrm{d}}} \end{cases} \tag{37}$$

The first term of each formula is designed to reduce the change rate of the desired point when the control error is being larger, so as to prevent the gap between the UAV state and the desired value from being too large. Define the Lyapunov function as:

$$V_r = \frac{1}{2}e_{r_{\mathrm{d}}}^2 + \frac{1}{2}e_{\alpha_{\mathrm{d}}}^2 + \frac{1}{2}e_{\beta_{\mathrm{d}}}^2 \tag{38}$$

Taking the derivative of (38) gives:

$$\begin{aligned} \dot{V}_r &= -e_{r_{\mathrm{d}}}\dot{r}_{\mathrm{d}} - e_{\alpha_{\mathrm{d}}}\dot{\alpha}_{\mathrm{d}} - e_{\beta_{\mathrm{d}}}\dot{\beta}_{\mathrm{d}} \\ &= -\frac{2e^{-c_1|r_1 - r_{\mathrm{d}}|}}{1+e^{-c_1|r_1 - r_{\mathrm{d}}|}}k_r e_{r_{\mathrm{d}}}^2 - \frac{2e^{-c_2|\alpha_1 - \alpha_{\mathrm{d}}|}}{1+e^{-c_2|\alpha_1 - \alpha_{\mathrm{d}}|}}k_\alpha e_{\alpha_{\mathrm{d}}}^2 \\ &\quad - \frac{2e^{-c_3|\beta_1 - \beta_{\mathrm{d}}|}}{1+e^{-c_3|\beta_1 - \beta_{\mathrm{d}}|}}k_\beta e_{\beta_{\mathrm{d}}}^2 \end{aligned} \tag{39}$$

The above equation is negative definite, and the errors will converge to zero over time. Then the desired position $\boldsymbol{p}_d^D$ in the coordinates $\{\boldsymbol{D}\}$ is:

$$\boldsymbol{p}_{\mathrm{d}}^D = \begin{bmatrix} r_{\mathrm{d}}\sin\alpha_{\mathrm{d}}\cos\beta_{\mathrm{d}} \\ r_{\mathrm{d}}\sin\alpha_{\mathrm{d}}\sin\beta_{\mathrm{d}} \\ r_{\mathrm{d}}\cos\alpha_{\mathrm{d}} \end{bmatrix} \tag{40}$$

Then the position error of the active docking UAV can be expressed as:

$$\boldsymbol{e}_p = \boldsymbol{R}_{B_1}^I \boldsymbol{d}_D^{B_1} + \boldsymbol{R}_D^I \boldsymbol{p}_{\mathrm{d}}^D \tag{41}$$

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Where, $\boldsymbol{d}_D^{B_1}$ is obtained from (5), $\boldsymbol{R}_D^I$ is the rotation matrix from the inertial frame to the contact frame.

After reaching the tracking point $\boldsymbol{p}_T$, the docking stage will not be entered immediately. The UAV will wait until the controller's estimation of the speed becomes stable, that is, the standard deviation of the estimated speed within the past 0.3 seconds is less than 0.1 m/s. At this time, the motion of the two UAVs is close to relative static, and the controller has a relatively accurate estimation of the speed of the passive docking UAV, then the docking process will enter the next phase.

C. Docking Phase

At this phase, the final desired position $\boldsymbol{p}_D$ should be reached to complete docking, and the relative velocity, relative attitude angle and relative angular velocity should be as small as possible. In fact, if the relative velocity of two UAVs is close within a period of time, the error of relative attitude angle and relative angular velocity will naturally be small. Here, a function $f(x)=\dfrac{1}{1+e^{-x}}$ is used to design the docking trajectory. When the horizontal coordinate tends to infinity, the function will approach 1 infinitely, and the first and second derivatives of the function are continuous and stable, which is in line with the requirements of the docking task. The function and the first and second derivatives are shown in Fig. 3.

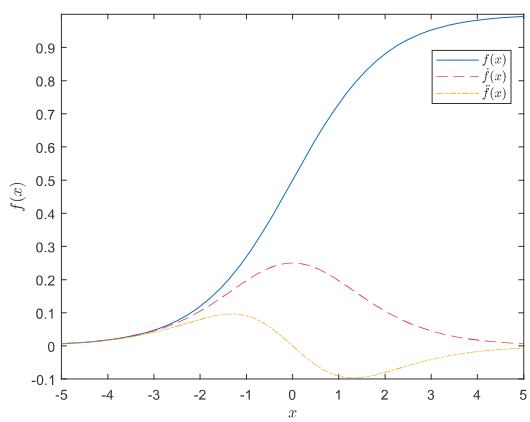

Fig. 3. Function f(x) and its first and second derivatives.

Let the horizontal coordinate of the function shift t_0 to the right and be scaled by a, and make the domain map to $[0,1)$, so that $g(0)=0$, $\lim_{t\to\infty} g(t)=d$. Thus, on the interval $[0,+\infty)$ we have:

$$g(t)=\left(\frac{1}{1+e^{-a(t-t_0)}}-\frac{1}{1+e^{at_0}}\right)\left(\frac{1}{e^{at_0}}+1\right)d \tag{42}$$

$$\max\left|\dot{g}(t)\right|=\frac{ad}{4}\left(\frac{1}{e^{at_0}}+1\right) \tag{43}$$

$$\max\left|\ddot{g}(t)\right|\approx 0.096a^2d\left(\frac{1}{e^{at_0}}+1\right) \tag{44}$$

From the point of view of the maximum first and second derivatives, the value of the parameter should not be too large to ensure the desired velocity and acceleration is small. Let:

$$r_{\mathrm{d}}=(1-g(t-t_{\mathrm{dock}}))d \tag{45}$$

Then, set α_d and β_d to zero and substitute (45) into (40) and (41) to get the position error.

V. Simulation Results

In order to verify the feasibility of the controller and the docking scheme, simulation experiments were carried out for verification and analysis. Some basic parameters in the experiment are given as follows: the mass of the UAV is $m=1.2\mathrm{kg}$, the gravitational acceleration is $g=9.8\mathrm{m/s}^2$, the inertia tensor of the drone is $\boldsymbol{J}=\mathrm{diag}(0.026,0.026,0.05)\mathrm{kg}\cdot\mathrm{m}^2$, Let $\boldsymbol{E}$ be 3×3 identity matrix, set control parameters as follows: $\boldsymbol{K}_1=\boldsymbol{E}$, $\boldsymbol{K}_2=6\boldsymbol{E}$, $\boldsymbol{\Gamma}=1.5\boldsymbol{E}$, $\boldsymbol{\Lambda}_{\boldsymbol{\alpha}}=2\boldsymbol{E}$, $\boldsymbol{\Lambda}=4\boldsymbol{E}$, $\boldsymbol{K}_3=0.5\boldsymbol{E}$, $\boldsymbol{K}_4=12\boldsymbol{E}$, $\boldsymbol{\Lambda}_\tau=0.2\boldsymbol{J}$, $\varepsilon=6$, $\Delta=0.2$, $\kappa_1=0.5$, $\kappa_2=5$, $k_r=1.5$, $k_\theta=0.2$, $k_\varphi=0.2$, $c_1=4$, $c_2=3$, $c_3=3$, $d=2$, $a=0.75$, $t_0=2.5$.

This docking method is suitable for docking with hovering or uniform moving UAVs. Considering docking with a hovering UAV is simpler, only the results of docking with moving UAV are shown here. For the passive docking UAV, the initial position is $(2,1,3)^T$ and initial speed is set as $(2,0,0)^T$. For the active docking UAV, the initial position is $(-2,-2,1)^T$. The UAV is subjected to unknown external disturbance $\boldsymbol{f}_\varepsilon=(0.25,0.25,0.25)^T$, $\boldsymbol{\tau}_\varepsilon=(0.2,-0.2,0.05)^T$. At the same time, gaussian white noise is added into the sensor, and the standard deviation of position measurement noise is set to be 0.02m, the standard deviation of velocity measurement noise is set to be 0.02m /s, the standard deviation of pitch and roll angle measurement noise is set to be 0.03°, and the standard deviation of yaw angle measurement noise is set to be 0.1°. The three-dimensional trajectories of the two UAVs are shown in Fig. 4, wherein UAV(A) and UAV(B) represent the active and passive docking UAV respectively.

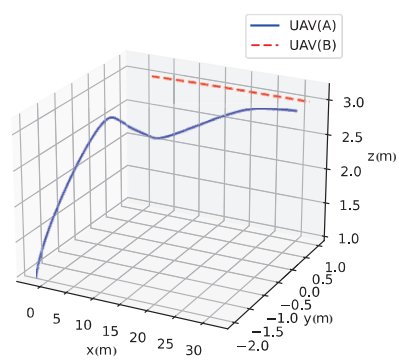

Fig. 4. Docking trajectories of UAVs.

The position errors, speed errors, attitude angle errors and angular velocity errors in docking process are shown in Fig. 5-8. It can be seen from the figures that two UAVs successfully dock and maintain small errors in speed, attitude angle and angular velocity.

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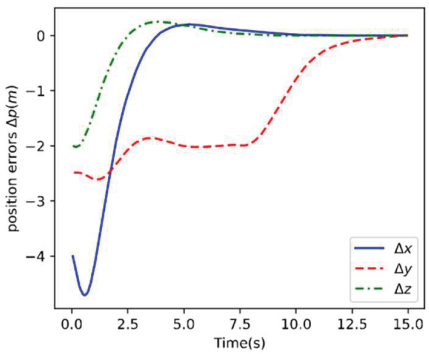

Fig. 5. Docking position errors.

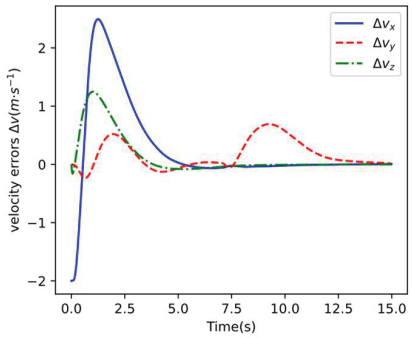

Fig. 6. Docking velocity errors.

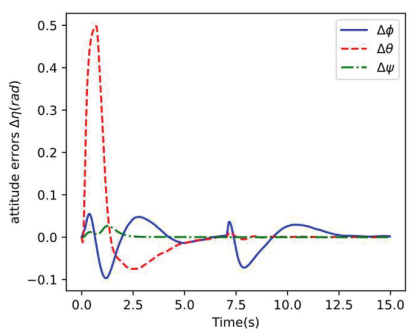

Fig. 7. Docking attitude angle errors.

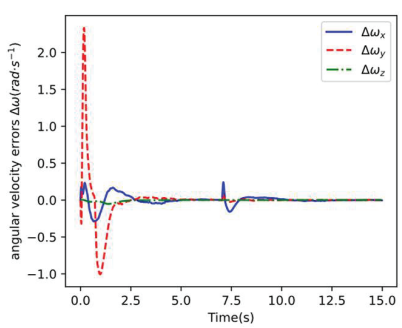

Fig. 8. Docking angular velocity errors.

The velocity of desired point estimated by controller is shown in Fig. 9. It can be seen that the estimated velocity of the desired point successfully converges with the speed of the passive docking UAV.

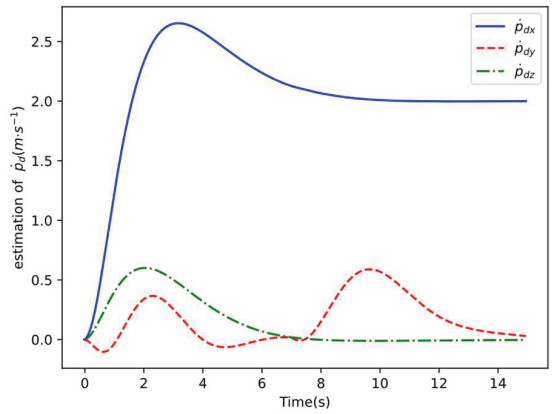

Fig. 9. Velocity estimation of desired point.

Due to the disturbance in the actual environment and the aerodynamic coupling effect between UAVs, velocity disturbance is added to the passive UAV, and the speed of the passive UAV is set as $(2+0.1\sin(0.8t),\ 0.2\sin(0.8t),0)^T$ m/s . Other conditions are the same. The three-dimensional trajectories of the two UAVs are shown in Fig. 10. The method in this paper is still valid when the motion is not completely uniform.

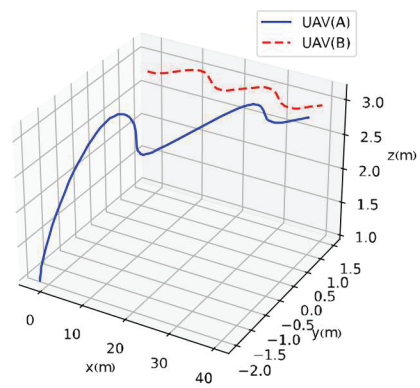

Fig. 10. Trajectories of UAVs when docking under velocity disturbance.

VI. Conclusion

(1) A controller for autonomous docking is proposed, which can estimate the speed of the desired point online and restrain the external disturbance, so that the active docking UAV can track the expected trajectory when only the expected position is given, and the attitude of the passive docking UAV is introduced at the docking end to reduce the attitude difference between the UAVs.

(2) The guidance method of UAV docking is designed, and the whole docking is completed in three phases: the approximation phase, the tracking phase and the docking phase. The trajectory generation method is specifically designed, so that the docking can be completed stably with small state error.

(3) Simulation results show that the designed docking method is suitable for the autonomous docking task of the quadrotor UAV. The method can complete the docking even under velocity disturbance.

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